Department of Information Science & Engineering
LateX LAB programs

1. Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

```
\documentclass{article}
```

```
% Header and Footer
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{} % Clear default header and footer
\rhead{Latex Lab} % Header - Title of the document
\cfoot{RNS Institute of Technology \quad \thepage} % Footer -
Institute name and page number

\begin{document}

        % Title
        \title{Test Document}
        \author{}
        \date{}
        \maketitle

        % Section 1
        \section{Section 1}
        This is a paragraph of text in Section 1.

        % Section 2
        \section{Section 2}
        This is another paragraph of text in Section 2.

\end{document}
```

Output

Test Document

1 Section 1

This is a paragraph of text in Section 1.

2 Section 2

This is another paragraph of text in Section 2.

2. Develop a LaTeX script to create a document that displays the sample Abstract/Summary

```
\documentclass{article}

\begin{document}

        % Title and Author
        \title{Sample Abstract}
        \author{Author Name}
        \date{} % Remove date
```

```
        \maketitle

        % Abstract Section
        \section*{Abstract}
        This is a sample abstract. Lorem ipsum dolor sit amet,
consectetur adipiscing elit. Sed ac velit sit amet lacus consequat
dignissim. Mauris tincidunt justo eget velit dictum, vel consectetur
libero consectetur. Nullam auctor ultricies eros, quis eleifend metus
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morbi tristique senectus et netus et malesuada fames ac turpis
egestas. Integer faucibus commodo orci, ac tempus sapien molestie nec.
Phasellus ut dui nisi. Nullam non erat nec turpis faucibus hendrerit.
Curabitur nec mauris sit amet ligula facilisis pharetra. Duis id dolor
vitae sem venenatis ultrices.

        % Additional Sections (if needed)
        % \section{Introduction}
        % \section{Methodology}
        % \section{Results}
        % \section{Discussion}
        % \section{Conclusion}

\end{document}
```
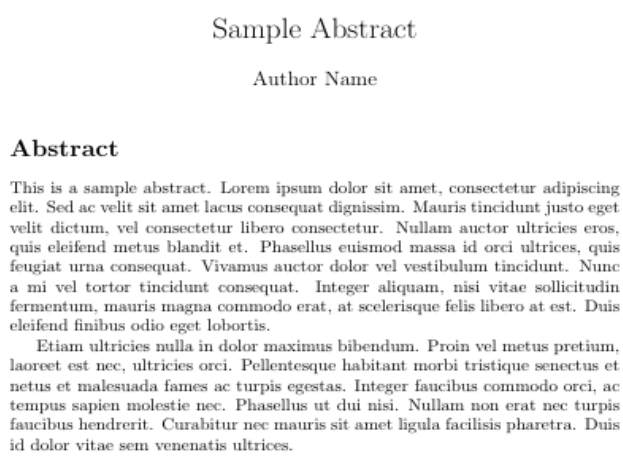

OUTPUT



Sample Abstract

Author Name

Abstract

This is a sample abstract. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed ac velit sit amet lacus consequat dignissim. Mauris tincidunt justo eget velit dictum, vel consectetur libero consectetur. Nullam auctor ultricies eros, quis eleifend metus blandit et. Phasellus euismod massa id orci ultrices, quis feugiat urna consequat. Vivamus auctor dolor vel vestibulum tincidunt. Nunc a mi vel tortor tincidunt consequat. Integer aliquam, nisi vitae sollicitudin fermentum, mauris magna commodo erat, at scelerisque felis libero at est. Duis eleifend finibus odio eget lobortis.

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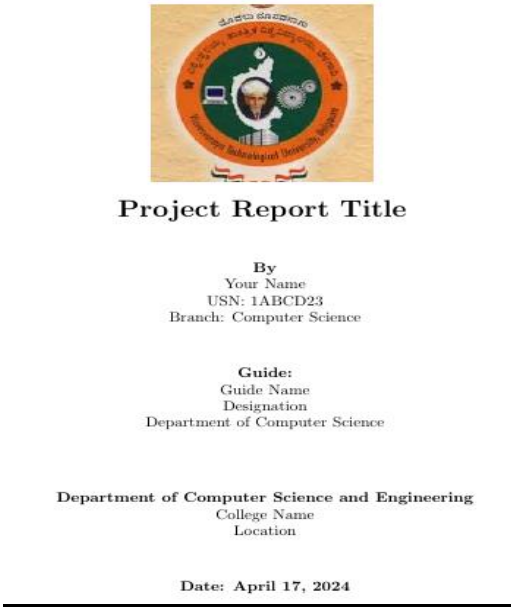
3. Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]

```
\documentclass{article}
\usepackage{graphicx}  % for including images
\usepackage{geometry}  % for setting page dimensions
```

```latex
\usepackage{setspace}  % for setting line spacing

% Define page dimensions
\geometry{a4paper, margin=1in}

\begin{document}

	\begin{titlepage}
		\centering

		% VTU Logo
		\includegraphics[width=0.3\textwidth]{vtu_logo.png}
		\vspace{0.5cm}

		% Title
		\textbf{\LARGE Project Report Title}
		\vspace{1cm}

		% Student Information
		\textbf{By}\\
		Your Name\\
		USN: 1ABCD23\\
		Branch: Computer Science\\
		\vspace{1cm}

		% Guide Information
		\textbf{Guide:}\\
		Guide Name\\
		Designation\\
		Department of Computer Science\\
		\vspace{1.5cm}

		% Institute Details
		\textbf{Department of Computer Science and Engineering}\\
		College Name\\
		Location\\
		\vspace{1cm}

		% Date
		\textbf{Date: \today} % Use today's date
	\end{titlepage}

\end{document}
```

<u>Output</u>



Project Report Title

By
Your Name
USN: 1ABCD23
Branch: Computer Science

Guide:
Guide Name
Designation
Department of Computer Science

Department of Computer Science and Engineering
College Name
Location

Date: April 17, 2024

4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry]

```
\documentclass[a4paper]{article}
\usepackage{graphicx}  % for including images
\usepackage{geometry}  % for setting page dimensions
\usepackage{setspace}  % for setting line spacing

% Define page dimensions
\geometry{margin=1in}

\begin{document}

        \thispagestyle{empty}  % Remove page number from the
certificate page

        % Title and Logo
        \begin{center}
                \includegraphics[width=0.3\textwidth]{vtu_logo.png}  %
University logo
                \vspace{1cm}

                \textbf{\Large Certificate of Completion}
                \vspace{0.5cm}
        \end{center}

        % Certificate Text
        \noindent
        This is to certify that \underline{\hspace{6cm}} (Registration
Number: \underline{\hspace{4cm}}) has successfully completed the
course on \underline{\hspace{8cm}} conducted by the Department of
\underline{\hspace{6cm}} at \underline{\hspace{8cm}}.
```

```
        \vspace{1cm}

        % Date
        \noindent
        Date: \underline{\hspace{4cm}}

        \vfill

        % Signature
        \begin{flushright}
                \textbf{Signature:} \underline{\hspace{6cm}}
        \end{flushright}

\end{document}
```

Output

Certificate of Completion

This is to certify that _____ (Registration Number: _____)
has successfully completed the course on _____ conducted
by the Department of _____ at _____.

Date: _____

5. Develop a LaTeX script to create a document that contains the following table with proper labels.

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

```
\documentclass{article}
\usepackage{array}  % for custom column formatting
\usepackage{booktabs}  % for better table rules
\usepackage{caption}  % for customizing table captions

\begin{document}

        \section*{Student Performance Table}
```

```
        \begin{table}[h]
                \centering
                \caption{Student Marks Summary}
                \label{tab:student-marks}
                \begin{tabular}{|c|c|l|c|c|c|}
                        \hline
                        \textbf{S.No} & \textbf{USN} & \textbf{Student
Name} & \textbf{Subject1} & \textbf{Subject2} & \textbf{Subject3} \\
                        \hline
                        1 & 4XX22XX001 & Name 1 & 79 & 60 & 90 \\
                        2 & 4XX22XX002 & Name 2 & 78 & 45 & 98 \\
                        3 & 4XX22XX003 & Name 3 & 36 & 75 & 59 \\
                        \hline
                \end{tabular}
        \end{table}

\end{document}
```

Output

Student Performance Table

Table 1: Student Marks Summary

S.No	USN	Student Name	Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	79	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	36	75	59

---

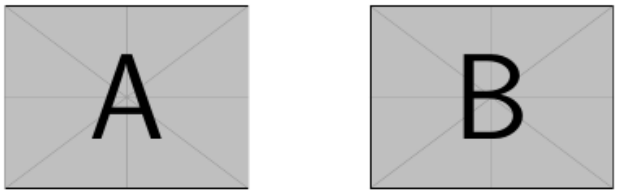
6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept

```
\documentclass{article}
\usepackage{graphicx}  % for including graphics
\usepackage{subcaption}  % for subfigures

\begin{document}

        \section{Side-by-Side Subfigures}

        \begin{figure}[htbp]
                \centering
                \begin{subfigure}[b]{0.4\textwidth}
                        \centering
                        \includegraphics[width=\textwidth]{example-image-
a}  % Include first image
                        \caption{Subfigure A}
                        \label{fig:subfig-a}
                \end{subfigure}
                \hfill
                \begin{subfigure}[b]{0.4\textwidth}
                        \centering
                        \includegraphics[width=\textwidth]{example-image-
b}  % Include second image
```

```
                    \caption{Subfigure B}
                    \label{fig:subfig-b}
            \end{subfigure}
            \caption{Two Side-by-Side Subfigures}
            \label{fig:side-by-side}
      \end{figure}

\end{document}
```

Output

1 Side-by-Side Subfigures



(a) Subfigure A (b) Subfigure B

Figure 1: Two Side-by-Side Subfigures